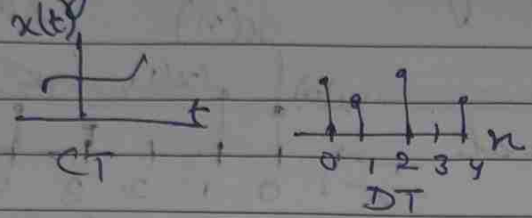
EIECC302 : Signals & Systems

1. Basics of signals & system
   Signals : Even/odd, periodic/non-periodic, CT/DT
   Systems : CTs / DTS, properties → Linearity
   → Time-invariant
   → Causality
   → Stability
   → with memory
   → Invertible



impulse func $\delta(t)$
unit-step $u(t)$
ramp $r(t)$

Transformation of independent variable

$$x(t) \longrightarrow x(t \pm t_0)$$
$$x(at)$$
$$x(-t)$$

LTI
$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$

2. LTI : convolution ⌈ convolution integral $\int$
   ⌊ convolution sum Σ

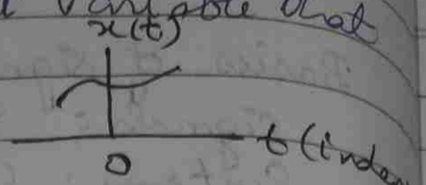
Fourier Analysis → Fourier Series (CT) → for periodic signals only.
→ DTFS

3. Fourier Transform (FT) : for non periodic signals

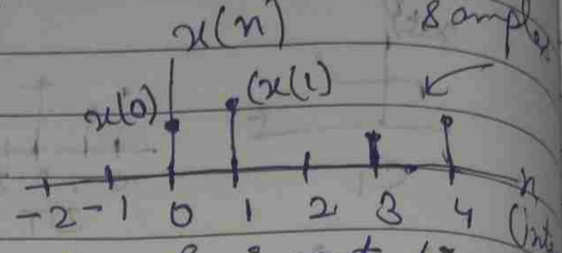
4. Laplace Transform (LT) $\int_{0^-}^{\infty}$ $\int_{-\infty}^{\infty}$
   Bilateral LT

5. Z-Transform – properties $\sum_{0}^{\infty}$ $\sum_{-\infty}^{\infty}$

Signal:
a function of independent variable that contains information.

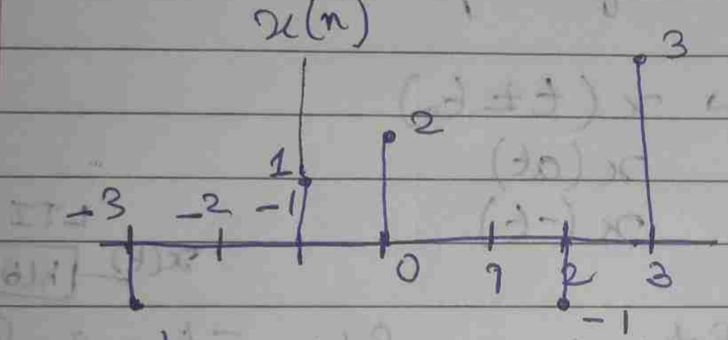


- Continuous-time signal (CT)
- Discrete time signal (DT)



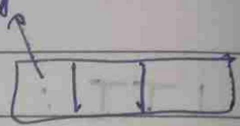
Sample is not discrete only time.

continuous-valued / Discrete-valued signal

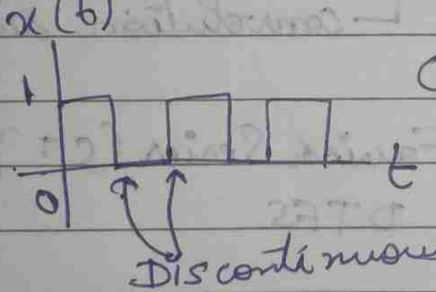


DT DV signal

signbits

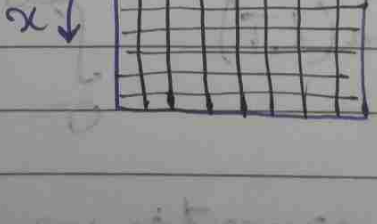


0 ∆ 0 0 → 0
0 ∆ 0 1 → 0.25
0 ∆ 1 0 → 0.5
0 ∆ 1 1 → 0.75



CT DV signal

Multichannel / Multidimensional signal



I(x, y)

000

111

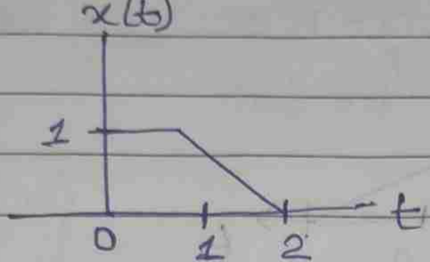
- No. of pixels
- No. of bits

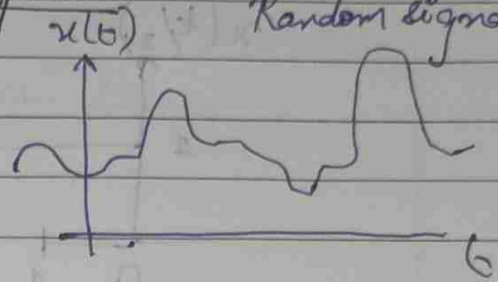
I(x, y, t)

ECG → Multi-channel, one-dimensional signal

$$I(x,y,t) = \begin{bmatrix} I_r(x,y,t) \\ I_g(x,y,t) \\ I_b(x,y,t) \end{bmatrix}$$

Deterministic / Random Signal:





Random signal

$$x(t) = \begin{cases} 0 & t<0 \\ 1 & 0<t<1 \\ 2-t & 1<t<2 \end{cases}$$

Transformation of Independent Variable

$x(t)$ — t: independent variable

$x(t \pm t_0)$ --- time-shifting

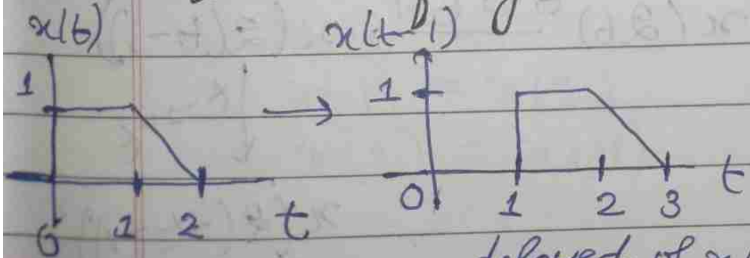
$x(at)$ --- time-scaling

$x(-t)$ --- time-reversal

$x(-at \pm t_0)$

$$x(t) = \begin{cases} 1 & 0<t<1 \\ 2-t & 1<t<2 \\ 0 & \text{otherwise.} \end{cases}$$

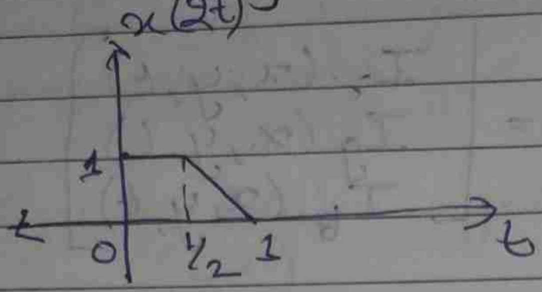
Time shifting



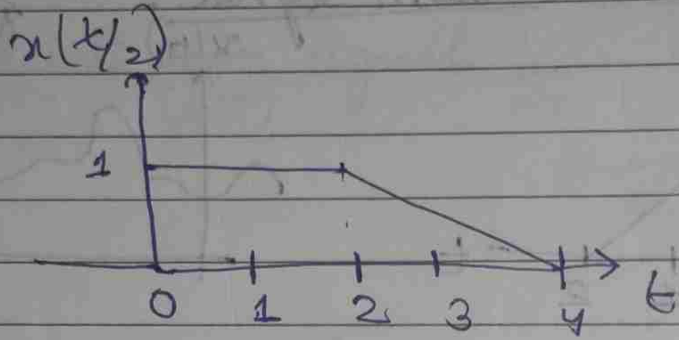
delayed of x(t)

$$x(t-1) = \begin{cases} 1 & 1<t<2 \\ 2-(t-1) = 3-t & 2<t<3 \\ 0 & \text{otherwise} \end{cases}$$

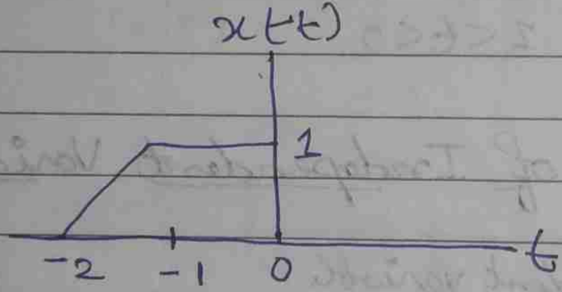
Time scaling



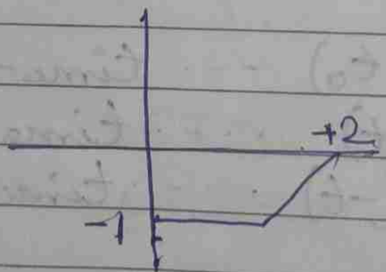
$$x(2t) = \begin{cases} 1 & 0<2t<1 \Rightarrow 0<t<\frac{1}{2} \\ 2-2t & \frac{1}{2}<t<1 \\ 0 & \text{otherwise} \end{cases}$$



Time-reversal:



→ $x(t)$



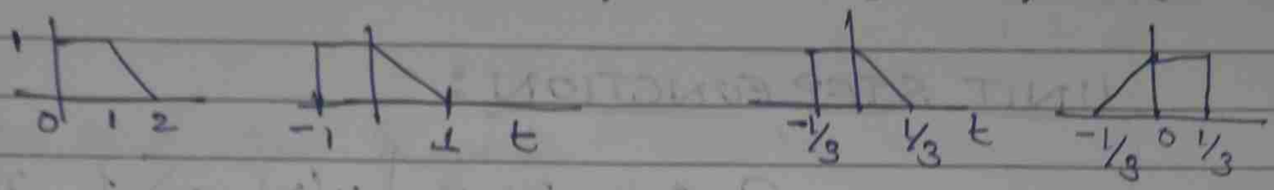
$$x(t) \xrightarrow{t \to t-1} x(t-1) \xrightarrow{t \to 2t} x(2t-1) \xrightarrow{t \to -t} x(-2t-1)$$

$$x(t) \xrightarrow{t \to 2t} x(2t) \xrightarrow{t \to t-1} x(2(t-1)) \xrightarrow{t \to -t} x(2(-t-1))$$

$x(-3t+1)$

eg

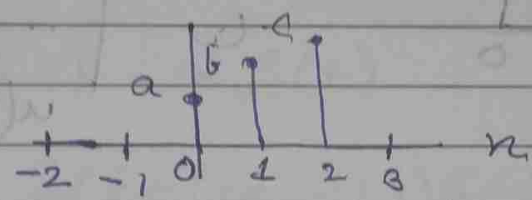
$x(t) \longrightarrow x(t+1) \longrightarrow x(3t+1) \longrightarrow x(-3t+1)$

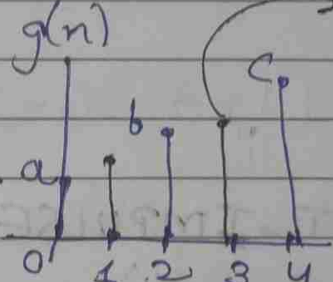


$x(n)$: $x(n \pm n_0)$ integer.

$x(-n)$

eg $x(n)$





interpolation (estimating the value b/w two known values)

$g(n) = x\left(\frac{n}{2}\right)$

$g(0) = x\left(\frac{0}{2}\right) = a$

$g(1) = x\left(\frac{1}{2}\right) = $ (not defined) $= 0$

$g(2) = x\left(\frac{2}{2}\right) = x_1$

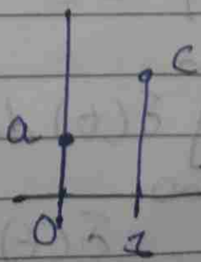
(oversampling)

$r(n) = x(2n)$

$r(0) = x(0)$

$r(1) = x(2)$

$r(2) = x(4)$



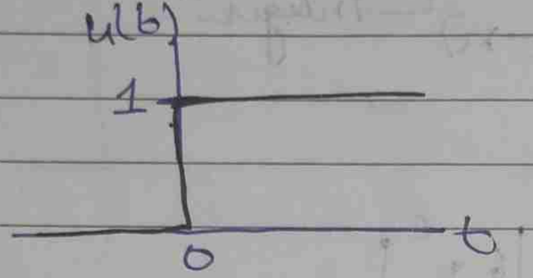
(undersampling)

Books

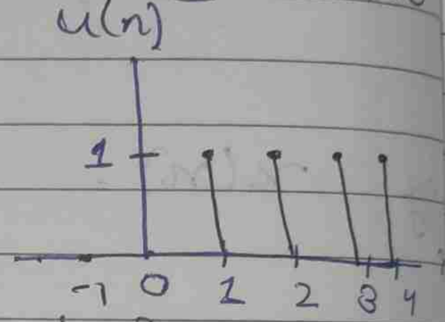
1. oppenheim (SnS)
2. Tarun Rawat (SnS)

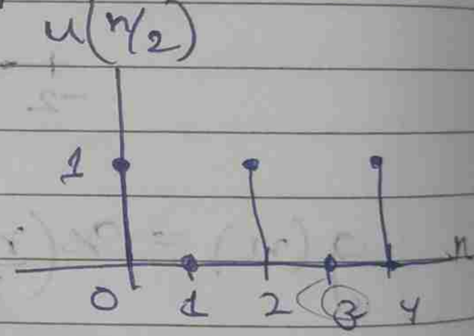
UNIT STEP FUNCTION :

$$u(t) = \begin{cases} 1 & t>0 \\ 0 & t<0 \end{cases}$$

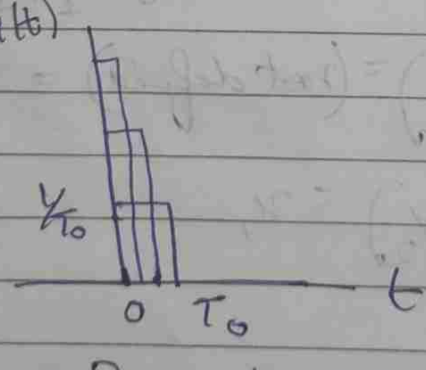


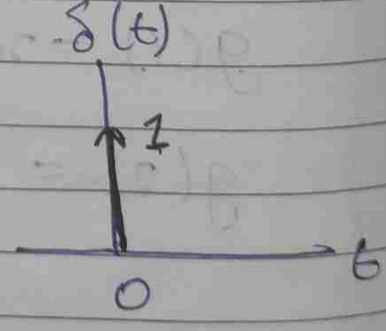
$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n<0 \end{cases}$$





UNIT IMPULSE FUNCTION



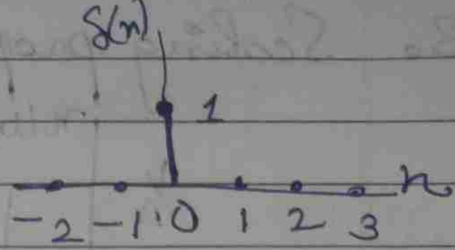


$$\delta(t) = 0 \quad t \neq 0$$

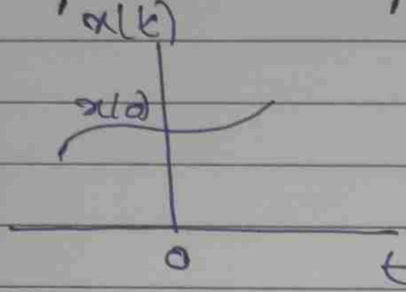
$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

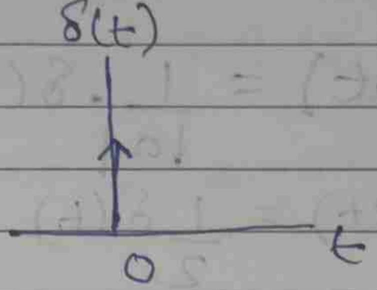
$$\delta(t) = \lim_{T_0 \to 0} x(t)$$

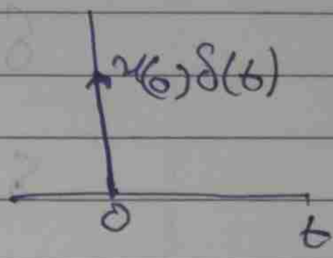
$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



1. Multiplication Property







$$x(t)\,\delta(t) = x_0\,\delta(t)$$

$$x(t)\,\delta(t-1) = x(1)\,\delta(t-1)$$

$$x(t)\,\delta(t+1) = x(-1)\,\delta(t+1)$$

$$x(t)\,\delta(t-t_0) = x(t)\Big|_{t=t_0} \cdot \delta(t-t_0)$$

2. Sifting property

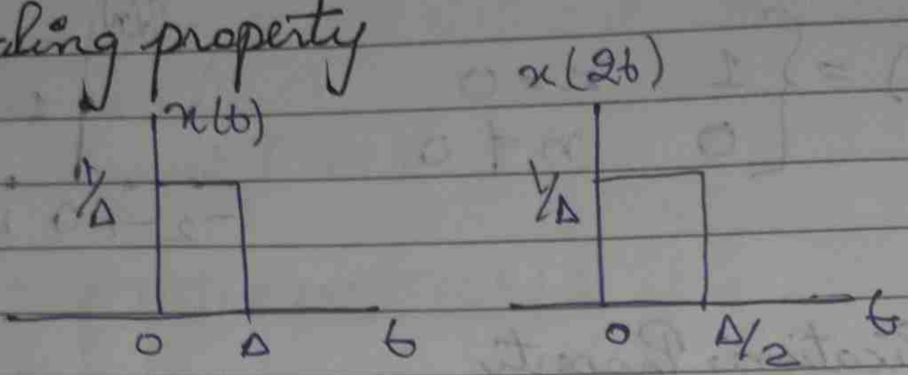
$$\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = \int_{-\infty}^{\infty} x(0)\,\delta(t)\,dt$$

$$= x(0)\underbrace{\int_{-\infty}^{\infty}\delta(t)\,dt}_{=1}$$

$$= x(0)$$

$$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t)\Big|_{t=t_0} = x(t_0)$$

3. Scaling property



$$\delta(at) = \frac{1}{|a|} \cdot \delta(t)$$

$$\delta(2t) = \frac{1}{2} \delta(t)$$

$$\delta(t/3) = 3\delta(t)$$

4 Even function

$$\delta(t) = \delta(-t)$$

eg
$$\int_{-\infty}^{\infty} t \cdot \delta(t-1)\, dt = t\Big|_{t=1} = 1$$

$$\int_{-1}^{3} t\,\delta(t-2)\, dt = t\Big|_{t=2} = 2$$

$$\int_{-1}^{3} t\,\delta(t+2)\, dt = 0$$

$$\int_{-\infty}^{\infty} \cos t\,\delta(t)\,dt = \cos(0) = 1$$

$$\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,\delta\tau =$$

$$= \int_{-\infty}^{\infty} x(t)\,\delta(t-\tau)\,d\tau$$

$$= x(t)\underbrace{\int_{-\infty}^{\infty} \delta(t-\tau)\,d\tau}_{=1}$$

$$= x(t)$$

$$\boxed{\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,\delta\tau = x(t)}$$

DISCRETE :

1. $x(n)\,\delta(n) = x(n)\big|_{n=0}\,\delta(n) = x(0)\,\delta(n)$

$$x(k)\,\delta(n-k) = x(n)\,\delta(n-k)$$

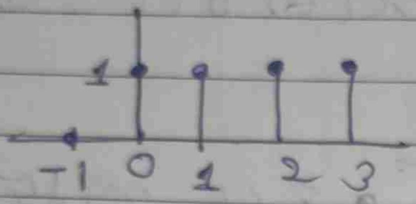
2. $$\delta(n) = \begin{cases} 1 & n=0 \\ 0 & n\neq 0 \end{cases}$$

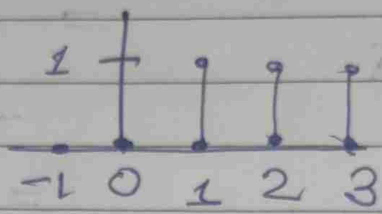
$$\delta(kn) = \begin{cases} 1 & kn=0 \Rightarrow n=0 \\ 0 & kn\neq 0 \Rightarrow n\neq 0 \end{cases} = \delta(n)$$

$$\delta(kn) = \delta(n)$$
$$\delta(2n) = \delta(n)$$

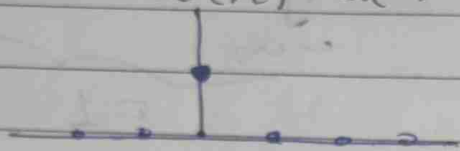
$\Rightarrow u(n) - u(n-1) = \delta(n)$

$u(n)$ $\qquad\qquad$ $u(n-1)$





$u(n) - u(n-1)$



$$\Rightarrow \sum_{k=-\infty}^{n} \delta(k) = u(n)$$

$$\sum_{k=-\infty}^{n} \delta(k) = \sum_{m=0}^{\infty} \delta(n-m) = \delta(n) + \delta(n-1) + \delta(n-2) + \cdots = u(n)$$

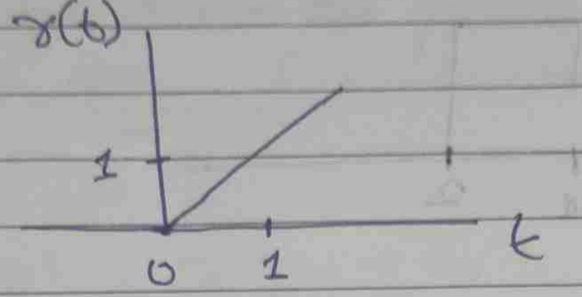
$k = n - m \Rightarrow m = n - k$

$k = -\infty \Rightarrow m = \infty$

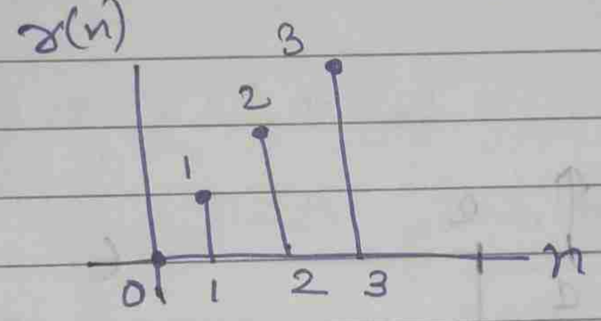
$k = n \Rightarrow m = 0$

UNIT RAMP FUNCTION

$$r(t) = t \cdot u(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$$r(n) = n\,u(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$



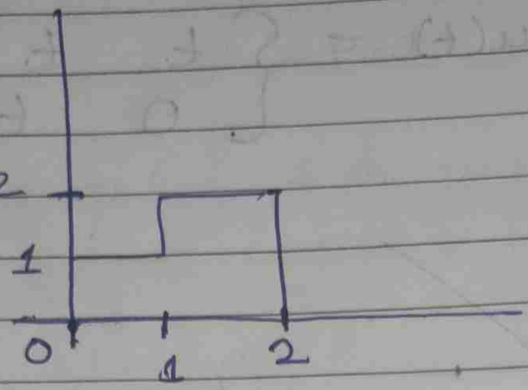
$$\boxed{r(t) \xrightarrow{d/dt} u(t) \xrightarrow{d/dt} \delta(t)}$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$$

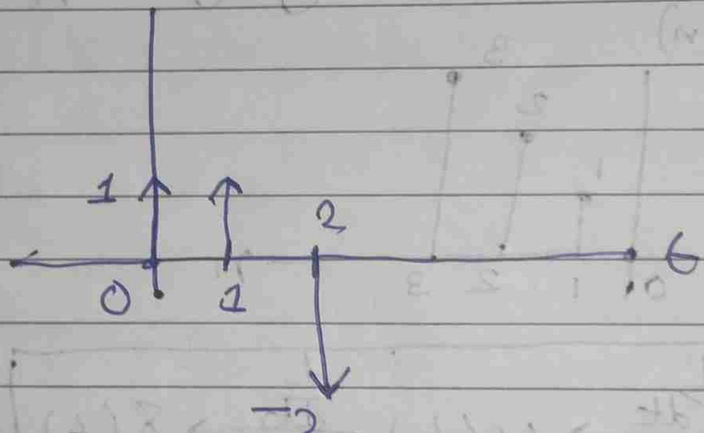
$$r(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$$

$$\boxed{\delta(t) \xrightarrow{\int(\,)d\tau} u(t) \xrightarrow{\int(\,)d\tau} r(t)}$$

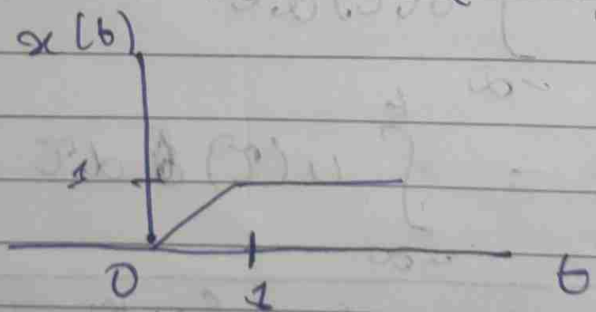
eg $x(t) = u(t) + u(t-1) - 2u(t-2)$



$$\frac{d}{dx} x(t) = \delta(t) + \delta(t-1) - 2\delta(t-2)$$



Ex $x(t) = r(t) - r(t-1)$



Periodic Signals

$$x(t) = x(t \pm T_0)$$

T_0 fundamental period

$\omega_0 = \frac{2\pi}{T_0}$ fundamental freq.

$$x(n) = x(n \pm N)$$

N fundamental period (integer)

eg $x(t) = e^{j\omega_0 t}$

$$x(t) = x(t + T_0)$$

$$= e^{j\omega_0 t} \cdot \underbrace{e^{j\omega_0 T_0}}_{=1 \text{ (should be 1)}}$$

$$e^{j\omega_0 T_0} = 1 = e^{j2m\pi}$$

$$\omega_0 T_0 = 2m\pi$$

$$T_0 = \frac{2\pi}{\omega_0} m$$

$m = 1$; $T_0 = \frac{2\pi}{\omega_0}$

eg $x(n) = e^{j\omega_0 n}$

$$x(n) = x(n + N)$$

$$= e^{j\omega_0 n} \cdot \underbrace{e^{j\omega_0 N}}_{=1}$$

$$e^{j\omega_0 N} = 1 = e^{j2m\pi}$$

$$\omega_0 N = 2\pi m$$

$$N = \frac{2\pi}{\omega_0} m$$

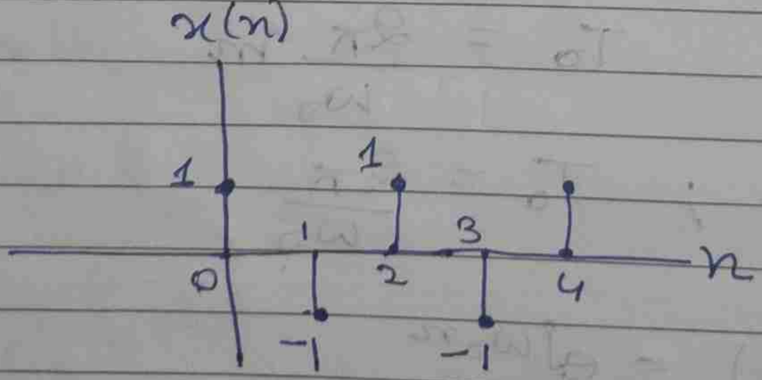
$$\boxed{\frac{\omega_0}{2\pi} = \frac{m}{N}}$$ rational no.

eg $$x(n) = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

$$\omega_0 = \pi$$

$$\frac{\omega_0}{2\pi} = \frac{1}{2} = \frac{m}{N}$$ rational no.

$\boxed{N = 2}$ 2 samples/cycles



$\boxed{f_d = 1/2 \text{ cycle / sample}}$ Max freq

$\boxed{\omega_d = \pi}$

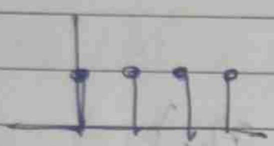
ex $x(n) = e^{j3\pi n} = (e^{j3\pi})^n = (-1)^n$

$\omega_0 = 3\pi$

$\frac{3\pi}{2\pi} = 3/2 = \frac{m}{N}$ = rational number

$\boxed{N = 2}$ samples/cycles

- Every discrete time signal is period in frequency domain with period 2π.



period with any arbitary integer greater than or equal to 2.

eg
$$x_1(t) = x_1(t + pT_1)$$
$$x_2(t) = x_2(t + qT_2)$$

$$x(t) = x_1(t) + x_2(t)$$

Is $x(t)$ periodic?

Solⁿ Let $x(t)$ be periodic with period T_0

$$x(t+T_0) = x_1(t+T_0) + x_0(t+T_0)$$

$$T_0 = pT_1 = qT_2$$

$$= x_1(t+pT_1) + x_2(t+qT_2)$$

$$x(t+T_0) = x_1(t) + x_2(t) = x(t)$$

$\frac{T_1}{T_2} = \frac{q}{p}$ rational no.

$\boxed{T_0 = pT_1}$

ex

$$x_1(t) = x_1(t+pT_1)$$
$$x_2(t) = x_2(t+qT_2)$$
$$x_3(t) = x_3(t+rT_3)$$

$$x(t+T_0) = x_1(t+T_0) + x_2(t+T_0) + x_3(t+T_0)$$

$$\frac{T_1}{T_2} = \frac{q}{p} = \frac{10}{8} = \frac{5}{4} = \frac{n_1}{n_2}$$

$$\frac{T_1}{T_2} = \frac{r}{p} = \frac{3}{8} = \frac{3}{8} = \frac{n_3}{n_4}$$

$$\boxed{T_0 = mT_1}$$

where $m = \text{lcm}(n_2, n_4)$

Q. 1.14, 1.15

(a) $x_1(t) = j e^{j10t}$

$$= e^{\pi/2} \cdot e^{j10t}$$
$$= e^{j10t + \pi/2}$$

$$e^{\pi/2} = \cos\pi/2 + j\sin\pi/2 = 0 + j$$

$$e^{j10t+\pi/2} = e^{j\omega_0 t + \phi}$$

$$\boxed{\omega_0 = 10}$$

Period $x(t) = x(t+T_0)$
$$= e^{j\omega_0 t} \cdot e^{j\omega_0 T_0}$$

$$e^{j\omega_0 T_0} = 1 = e^{2m\pi}$$

$$\omega_0 T_0 = 2m\pi$$

$$T_0 = \frac{2m\pi}{10}$$

$$= \frac{\pi}{5} m$$

$$\boxed{T_0 = \pi/5}$$

(b) $x_2(t) = e^{(-1+j)t}$

$$e^{(-1+j)t} = e^{j\omega_0 t}$$

$$\Rightarrow (-1+j)t = j\omega_0 t$$

$$\Rightarrow \omega_0 = \frac{-1+j}{j}$$

$$= -j(-1+j)$$

$$= j+1$$

$\therefore$ ω_0 is complex

$\Rightarrow$ $x_2(t)$ is non periodic.

(c) $x_3(n) = e^{j7\pi n}$

$$e^{j7\pi n} = e^{j\omega_0 n}$$

$$\cancel{7\pi n = \omega_0 N} \quad \boxed{\omega_0 = 7\pi}$$

$$\cancel{\frac{N}{m} = \frac{7\pi}{\,}}$$

$$\omega_0 N = 2m\pi$$

$$\frac{N}{m} = \frac{2\pi}{\omega_0} = \frac{2\pi}{7\pi} = \frac{2}{7} = \text{rational}$$

periodic signal $\boxed{N=2}$

(d) $x_4(n) = 3e^{j3\pi(n+1/2)/5}$

$$= 3e^{j3\pi/10} e^{j3\pi/5 n}$$

$$= Ke^{j3\pi/5 n}$$

$$\omega_0 = 3\pi/5$$

$$\omega_0 N = 2m\pi$$

$$3\pi/5 \, N = 2\pi m$$

$$\frac{N}{m} = \frac{10}{3} = \text{rational no.}$$

$\boxed{N=10}$

$x_4(n)$ is periodic.

(e) $x_5(n) = 3e^{j3/5(n+1/2)}$

$$= Ke^{j3/5 n}$$

$$\omega_0 = 3/5$$

$$\frac{m}{N} = \frac{\omega_0}{2\pi} = \frac{3/5}{2\pi} = \frac{3}{10\pi} = \text{irrational no.}$$

$\Rightarrow x_5(n)$ is not periodic.

1.15 (a) $x_1(t) \Rightarrow$ not periodic $\Rightarrow x_1(t) = 0 \ (t<0)$

(b) $x_2(n) = u(n) + u(-n)$

$x_2(n+N) = u(n+N) = u(-n-N)$

Not periodic.

(c) $x_3(n) = \sum_{k=-\infty}^{\infty} [\delta(n-4k) - \delta(n-1-4k)]$

$$x_3(n+N) = \sum_{k=-\infty}^{\infty} [\delta(n+N-4k) - \delta(n+N-1-4k)]$$

$$= \sum_{k=-\infty}^{\infty} \left[\delta\left(n-4\left(k-\frac{N}{4}\right)\right) - \delta\left(n-1-4\left(k-\frac{N}{4}\right)\right)\right]$$

Let $= k - \frac{N}{4} = m$ where m is an integer

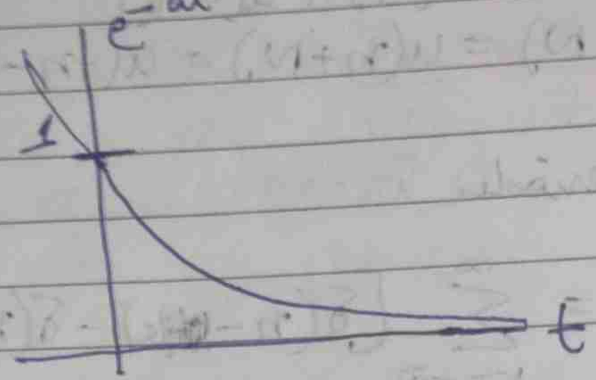
$$= \sum_{k=-\infty}^{\infty} [\delta(n-4m) - \delta(n-1-4m)]$$

$= x_3(n)$

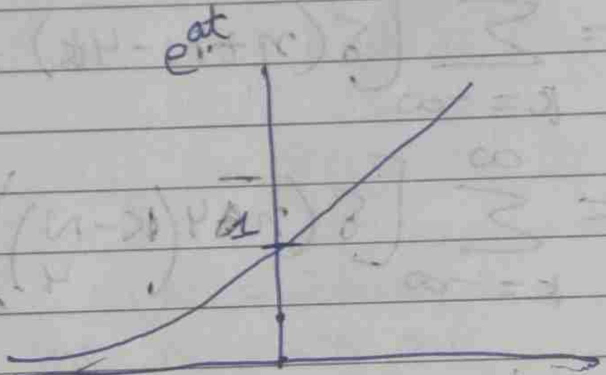
So, $x_3(n)$ is periodic when m is an integer.

Exponential Signals

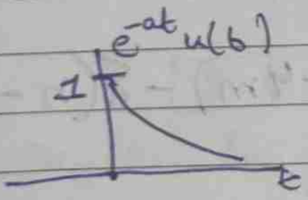
e^{-at}



e^{at}

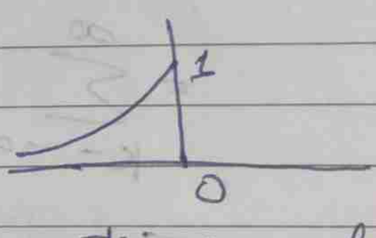


$e^{-at}u(t)$



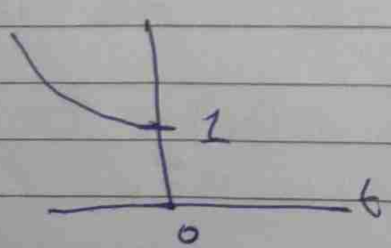
Causal signal
(right-sided)

$e^{+at}u(-t)$

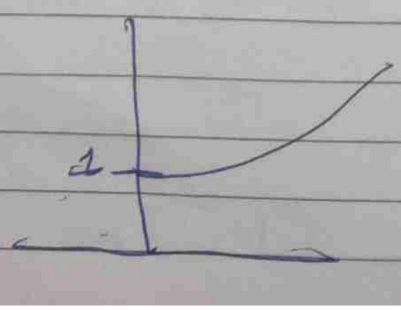


anti-causal signal
(left sided)

$e^{-at}u(-t)$

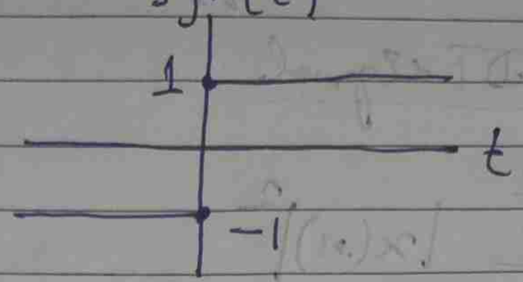


$e^{at}u(t)$



Signum function

$$\mathrm{sgn}(t) = \begin{cases} 1 & t>0 \\ -1 & t<0 \end{cases}$$

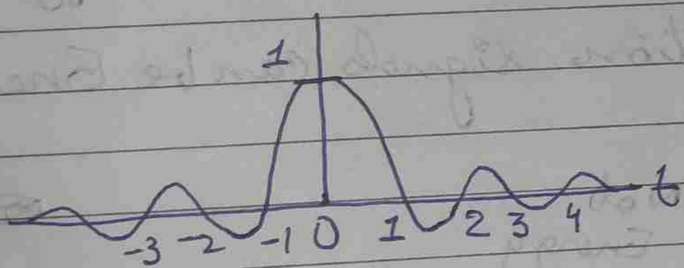


$$1 + \mathrm{sgn}(t) = 2u(t)$$

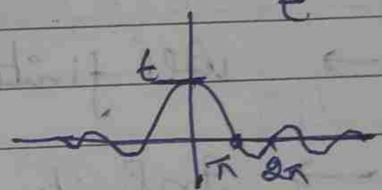
$$\Rightarrow \boxed{u(t) = \frac{1}{2}\left(1 + \mathrm{sgn}(t)\right)}$$

Sinc function

$$\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$



$$Sa(t) = \frac{\sin t}{t}$$



$\sin t = 0 = \sin m\pi$

$m = \pm 1, \pm 2, \ldots$

scaling this by π we get sinc fun

Energy & Power Signals

$x(t)$ .... CT signal

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

$$P_x = \lim_{t \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

OR

$$P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$x(n)$ ... DT signal

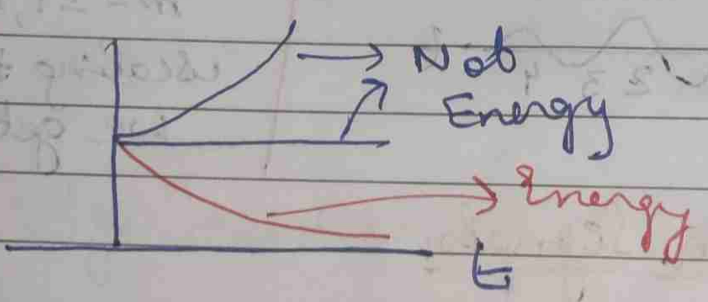
$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$P_x = \lim_{N \to \infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} |x(n)|^2$$

$0 < E_x < \infty$, $P = 0$ Energy Signal
$0 < P_x < \infty$, $E_x = \infty$ Power Signal
$E_x = \infty$, $P = \infty$ Neither energy nor Power.

→ All finite duration signals are Energy Signals

→ Infinite duration signals can be Energy or Power.



→ All periodic signals are Power signal but vice versa may not be true.

Ex $x(t)$; E_x

$kx(t)$	$k^2 E_x$
$x(-t)$	E_x
$x(t-t_0)$	E_x
$x(at)$	$\frac{E_x}{a}$
$x(at-b)$	$\frac{E_x}{a}$

$g(t) = x(at)$

$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$

$E_g = \int_{-\infty}^{\infty} x(at)^2 dt$

$at = \tau$

$dt = \frac{d\tau}{a}$

$= \frac{1}{a}\int_{-\infty}^{\infty} |x(\tau)|^2 d\tau$

$= \frac{E_x}{a}$

Ex $x(t) = A_c \cos \omega_c t$

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \cos^2 \omega_c t \, A_c^2 \, dt$$

$$= \frac{1}{2T}\int_{-T}^{T} \frac{A_c^2}{2}(1+\cos 2\omega_c t)\, dt$$

$$= \frac{1}{2T}\frac{A_c^2}{2}\left[T + \left(\frac{\sin 2\omega_c t}{2}\right)\right]_{-T}^{T}$$

$[2T + 0]$

$$= \frac{A_c^2}{2}$$

$\boxed{P_x = \frac{A_c^2}{2}}$ = Mean Square value.

$rms = \frac{A_c}{\sqrt{2}}$

Ex 1.19 Pg 47

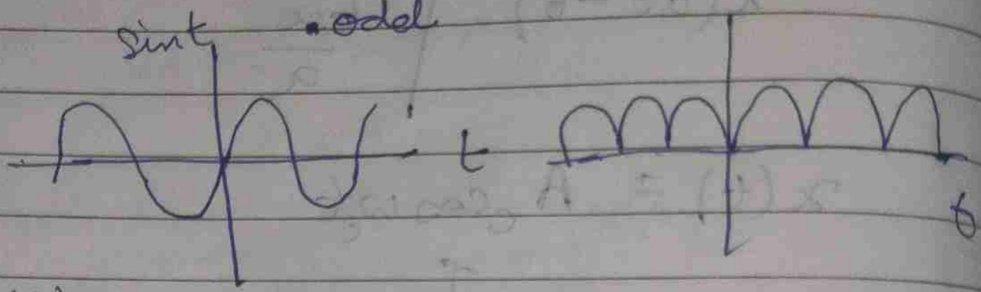
Even & Odd Signals

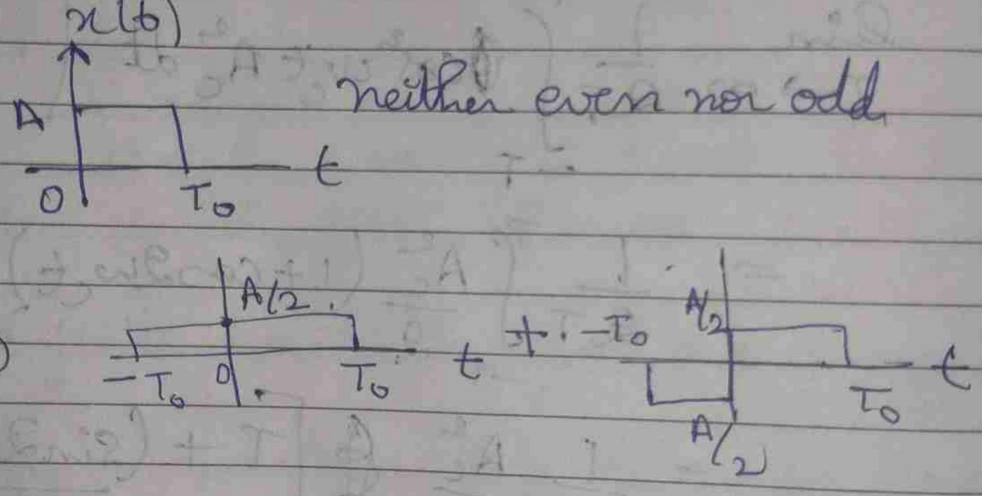
$x(t) = x(-t)$; $x(n) = x(-n)$ ∴ Even signal

$x(t) = -x(-t)$; $x(n) = -x(-n)$ Odd signal

or

$-x(t) = x(-t)$





neither even nor odd

$x(t) = -x(-t)$

$$\frac{dx(t)}{dt} = \frac{dx(-t)}{dt}$$

$g(t) = g(-t)$

$x(t)$ ---- neither even nor odd

$x(t) = x_e(t) + x_o(t)$ — ① where $x_e(t) = x_e(-t)$

$x(-t) = x_e(-t) + x_o(-t)$ $\quad x_o(t) = -x_o(-t)$

$= x_e(t) - x_o(t)$ — ②

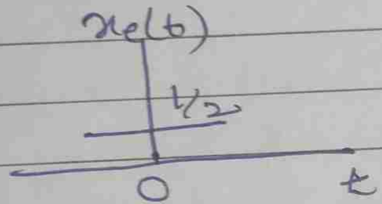
Add (1) & (2)

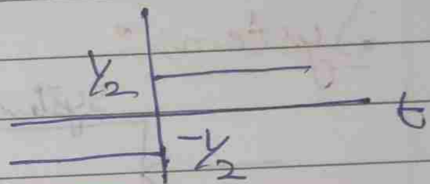
$$\left\{\begin{aligned} x_e(t) &= \frac{1}{2}\left[x(t) + x(-t)\right] \\ x_o(t) &= \frac{1}{2}\left[x(t) - x(-t)\right] \end{aligned}\right.$$

eg

$$x(t) = u(t)$$

$x_e(t)$ $\qquad\qquad$ $x_o(t) = \frac{1}{2}\operatorname{sgn}(t)$





Ex

$$\boxed{\int_{-\infty}^{\infty} x^2(t)\,dt = \int_{-\infty}^{\infty} x_e^2(t)\,dt + \int_{-\infty}^{\infty} x_o^2(t)\,dt}$$

$$x(t) = x_e(t) + x_o(t)$$

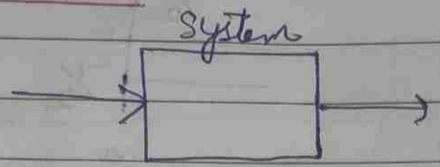
$$\int_{-\infty}^{\infty} x^2(t)\,dt = \int_{-\infty}^{\infty} \left[x_e(t) + x_o(t)\right]^2 dt,$$

$$\sum_{n=-\infty}^{\infty} x(n) = 0 \quad \text{- - if } x(n) \text{ is odd}$$

$$\sum_{n=-\infty}^{\infty} x(n) = 2\sum_{n=1}^{\infty} x(n) + x(0)$$

$$\sum_{n=-\infty}^{\infty} x^2(n) = \sum_{n=-\infty}^{\infty} x_e^2(n) + \sum_{n=-\infty}^{\infty} x_o^2(n)$$

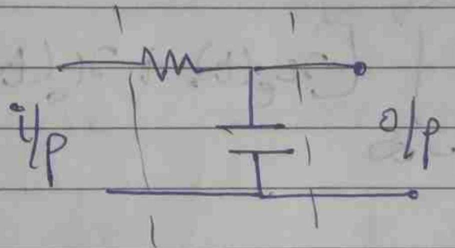
Systems:



$x(t) \longrightarrow \boxed{CTS} \longrightarrow y(t)$ $\qquad$ $x(n) \longrightarrow \boxed{DTS} \longrightarrow y(n)$

$x(t) \longrightarrow y(t)$ $\qquad$ $x(n) \longrightarrow y(n)$

e.g



SYSTEM PROPERTIES

1. LINEARITY :

A system is said to be linear if it follows the principle of superposition & homogenity.

$x_1(t) \longrightarrow y_1(t)$

$a\,x_1(t) \longrightarrow a\,y_1(t)$ - - Homogenous

$$x_2(t) \longrightarrow y_2(t)$$
$$b x_2(t) \longrightarrow b y_2(t)$$

$$a x_1(t) + b x_2(t) \longrightarrow a y_1(t) + b y_2(t)$$
... Linearity

Discrete

$$a x_1(n) + b x_2(n) \longrightarrow a y_1(n) + b y_2(n)$$

ex ⇒ $y(t) = t \cdot x(t)$

Linear?

$$y_1(t) = t \cdot x_1(t)$$
$$y_2(t) = t \cdot x_3(t)$$
$$y_3(t) = t \cdot x_3(t)$$

Let $x_3(t) = a x_1(t) + b x_2(t)$

For linearity

$$y_3(t) = a y_1(t) + b y_2(t)$$

LHS

$$\begin{aligned} y_3(t) &= t x_3(t) \\ &= t[a x_1(t) + b x_2(t)] \\ &= a t x_1(t) + b t x_2(t) \\ &= a y_1(t) + b y_2(t) \\ &= \text{RHS} \end{aligned}$$

∴ System is linear.

ex $y(t) = x^2(t)$ --- Non-linear

(b) $y(n) = 2x(n) - 3$

Non-relaxed

$y_1(n) = 2x_1(n) - 3$

$\therefore$ Non linear

(c) $y(t) = x(2t)$ .... Linear

(d) $y(t) = \mathrm{Re}\{x(t)\}$ ... Non linear

Let $x_1(t) = a + jb$

$y_1(t) = a$

$x_2(t) = j\,x_1(t) = ja - b$

$y_2(t) = -b \neq j\,y_1(t)$

(2) Time - Invariant

The behaviour of system does not change with time.

$$x(t) \longrightarrow y(t)$$
$$x(t - t_0) \longrightarrow y(t - t_0)$$

eg $y(t) = t \cdot x(t)$

$y_1(t) = t \cdot x_1(t)$

$y_2(t) = t \cdot x_2(t)$

Let $x_2(t) = x_1(t - t_0)$

$y_2(t) = y_1(t - t_0)$ ... for TI

LHS = $y_2(t) = t \cdot x_1(t - t_0)$

RHS $y_1(t-t_0) = ?$

$$y_1(t) = t\,x_1(t)$$
$$y_1(t-t_0) = (t-t_0)\,x_1(t-t_0)$$
$$\neq LHS$$

$\therefore$ TV

eg (a) $y(t) = x(t-1)$ ---- TI

(b) $y(t) = x(at)$ ---- TV

$$y_1(t) = x_1(at)$$
$$y_2(t) = x_2(at)$$

Let $x_2(t) = x_1(t-t_0)$

$$\Rightarrow x_2(at) = x_1(at-t_0)$$

$$y_2(t) = y_1(t-t_0) \quad \text{for TI}$$

$$LHS = y_2(t)$$
$$= x_2(at)$$
$$= x_1(at-t_0)$$

$$RHS = y_1(t-t_0)$$

$$y_1(t) = x_1(at)$$
$$y_1(t-t_0) = x_1(a(t-t_0))$$
$$\neq LHS$$

TV

(c) $y(t) = \sin[x(t)]$ ... TI

(d) $y(n) = x(-n)$ ... TV

(e) $y(t) = x(t) \cdot u(t)$ ... Linearity ✓ TI ✗

3. Causality:
A system is said to be causal if it does not anticipate with future inp

- All physically realisable system are cau

ex (a) $y(n) = n x(n)$ --- causal

(b) $y(t) = x(-t)$ --- Noncausal

$y(1) = x(-1)$

$y(-2) = x(2)$

(c) $y(n) = a x(n) + b x(n+1)$ --- non causal

4. Stable Systems:

BIBO $\Rightarrow$ Bounded i/p Bounded o/p.

$|x(t)| \le B < \infty$

$-B \le x(t) \le B$ --- Bounded input

$|y(t)| \le B_y < \infty$ --- Bounded o/p ex ⇒ $y(t) = t \cdot x(t)$

$|x(t)| \le B < \infty$ . . . Bounded i/p

$-B \le x(t) \le B$

$-tB \le y(t) \le tB$ . . . unbounded o/p

—— Non stable.

ex ⇒ $y(t) = e^{x(t)}$

$|x(t)| < B < \infty$ . . . Bounded i/p

$e^{-B} \le y(t) \le e^{B}$ . . . Bounded o/p

∴ stable.

eg (a) $y(n) = r^n x(n)$ range of r for stable?

(b) $y(n) = \sum_{k=0}^{\infty} \rho^k x(n-k)$